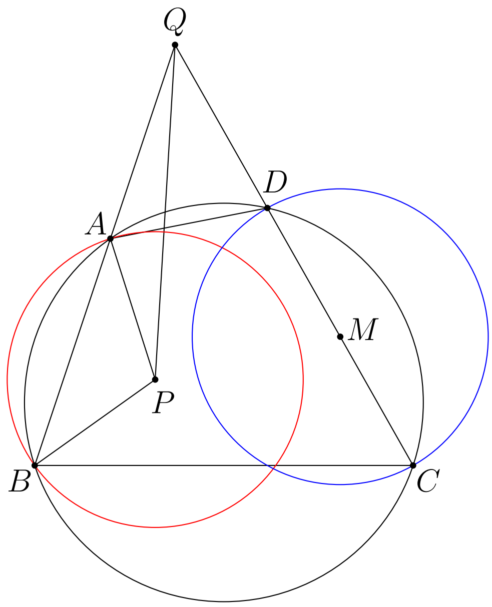
Power of a Point — Solutions

Problem 1. (Baltic Way 2016 Problem 20) Let $ABCD$ be a cyclic quadrilateral with AB and CD not parallel. Let M be the midpoint of CD. Let P be a point inside $ABCD$ such that $PA = PB = CM$. Prove that AB, CD and the perpendicular bisector of MP are concurrent.



Solution. Let Q be the intersection point of AB and CD. The power of Q with respect to the circle with centre P and radius $PA = PB$ is

$$QA \times QB = QP^2 - PA^2.$$

The power of Q with respect to the circle with centre M and radius $MC = MD$ is

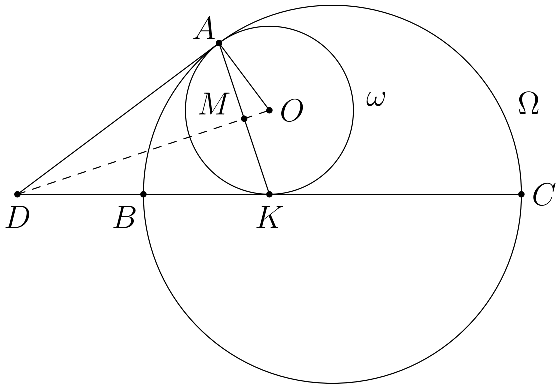
$$QD \times QC = QM^2 - MC^2.$$

It follows that

$$QP^2 = QA \times QB + PA^2 = QD \times QC + MC^2 = QM^2.$$

So $QP = QM$. Equivalently, AB, CD and the perpendicular bisector of MP are concurrent. $\square$

Problem 2. (Sharygin Geometry Olympiad 2021 Correspondence Problem 16) Let circles Ω and ω touch internally at point A. A chord BC of Ω touches ω at point K. Let O be the centre of ω. Prove that the circle BOC bisects segment AK.

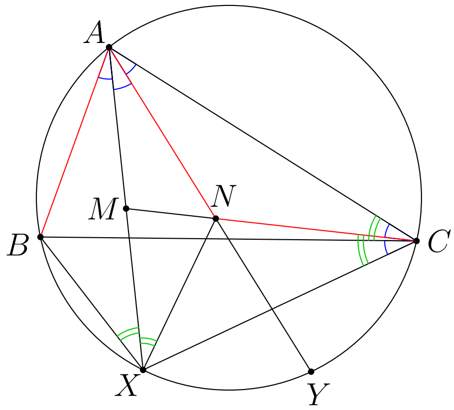


Solution. Let M be the midpoint of AK. Let the common tangent at A meet the line BC at D. Note that D, M, O are collinear since all of them lie on the perpendicular bisector of AK. Also, note that $\angle DAO = \angle DMA = 90°$, so we have $DM \times DO = DA^2$. It follows that

$$DM \times DO = DA^2 = DB \times DC.$$

This implies B, C, O, M are concyclic. $\square$

Problem 3. (European Mathematical Cup 2021 Junior Problem 2) Let ABC be an acute-angled triangle such that $AB < AC$. Let X and Y be points on the minor arc BC of the circumcircle of ABC such that $BX = XY = YC$. Suppose that there exists a point N on the segment AY such that $AB = AN = NC$. Prove that the line NC passes through the midpoint of the segment AX.



Solution. Let M be the intersection point of CN and AX. Firstly, the given condition implies $\angle BAX = \angle XAY = \angle YAC$. Since $NA = NC$, we have

$$\angle NCA = \angle NAC = \angle MAN.$$

Therefore, MA is tangent to (ANC), so that

$$MA^2 = MN \times MC.$$

Next, note that $\triangle ABX \cong \triangle ANX$ since $AB = AN$, $\angle BAX = \angle NAX$ and $AX = AX$. This implies

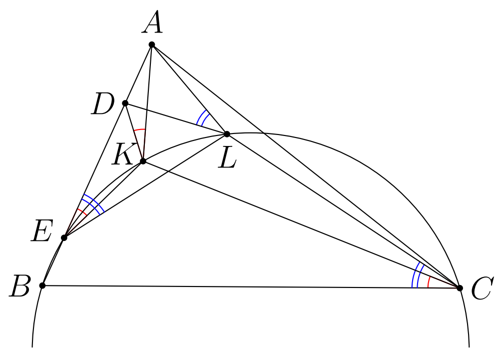
$$\angle AXN = \angle AXB = \angle ACB = \angle ACM + \angle MCB = \angle BCX + \angle MCB = \angle MCX.$$

Therefore, MX is tangent to (XNC), so that

$$MX^2 = MN \times MC = MA^2.$$

This shows M is the midpoint of AX. $\square$

Problem 4. (CGMO 2022 Problem 5) Points K and L are in the interior of a triangle ABC. Point D lies on the side AB. It is known that points B, K, L, C are concyclic, and $\angle AKD = \angle BCK$, $\angle ALD = \angle BCL$. Prove that $AK = AL$.



Solution. Suppose (BCK) meets AB again at E. From

$$\angle AKD = \angle BCK = \angle AEK,$$

we see that AK is tangent to (DEK). Similarly, from

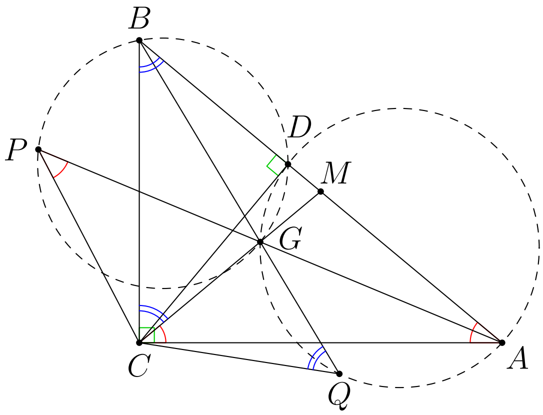
$$\angle ALD = \angle BCL = \angle AEL,$$

we see that AL is tangent to (DEL). It follows that

$$AK^2 = AD \times AE = AL^2,$$

so $AK = AL$. $\square$

Problem 5. (Canada MO 2013 Problem 3) Let G be the centroid of a right-angled triangle ABC with $\angle BCA = 90^\circ$. Let P be the point on ray AG such that $\angle CPA = \angle CAB$, and let Q be the point on ray BG such that $\angle CQB = \angle ABC$. Prove that the circumcircles of triangles AQG and BPG meet at a point on side AB.



Solution. Let M be the midpoint of AB, and let D be the foot of altitude of $\triangle ABC$ from C. Since $\angle ACB = 90^\circ$ and $MA = MB$, we have $MA = MB = MC$. This implies

$$\angle ACM = \angle MAC = \angle CPA.$$

Thus, AC is tangent to (CPG). This implies

$$AG \times AP = AC^2 = AD \times AB.$$

This shows D lies on (BPG). Similarly, from

$$\angle BCM = \angle MBC = \angle CQB,$$

we see that BC is tangent to (CQG). This implies

$$BG \times BQ = BC^2 = BD \times BA,$$

and hence D also lies on (AQG). Thus, (BPG) meets (AQG) again at D, which lies on AB. $\square$

Problem 6. (China High School Mathematics League Second Round 2021 Problem 2) In $\triangle ABC$, M is the midpoint of side AC. D, E are two points on the tangent line at A to the circumcircle of $\triangle ABC$, such that $MD//AB$, and A is the midpoint of DE. The circle passing through points A, B, E intersects the side AC again at P. The circle passing through points A, D, P intersects the extension of line DM again at Q. Prove that $\angle BCQ = \angle BAC$.

Solution. Let N be the midpoint of BC. By the midpoint theorem, we know that N lies on MD. As $\angle BAC = \angle QMC$, it suffices to show NC is tangent to (CQM). Firstly, as

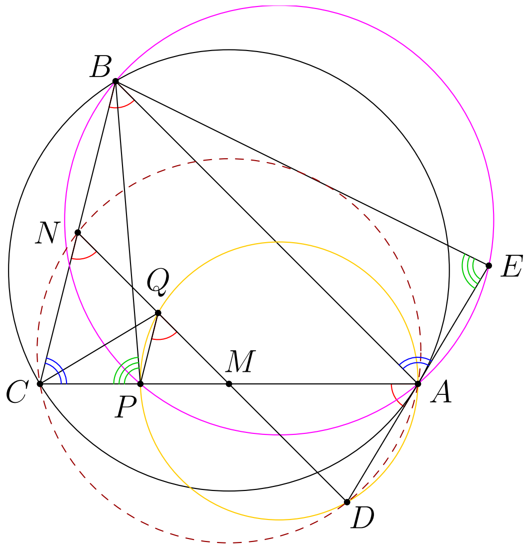
$$\angle CAD = \angle CBA = \angle CND,$$

the points C, D, A, N are concyclic. This implies $\triangle MNC \sim \triangle MAD$, and so

$$\frac{MN}{NC} = \frac{MA}{AD}. \tag{1}$$

Secondly, by Reim's theorem (or $\angle PQD = \angle PAD = \angle CND$), we have $QP//NC$. Therefore, we have

$$\frac{NQ}{NM} = \frac{CP}{CM}. \tag{2}$$



Thirdly, note that $\triangle BCP \sim \triangle BAE$ since $\angle BCP = \angle BAE$ and $\angle BPC = \angle BEA$. This implies

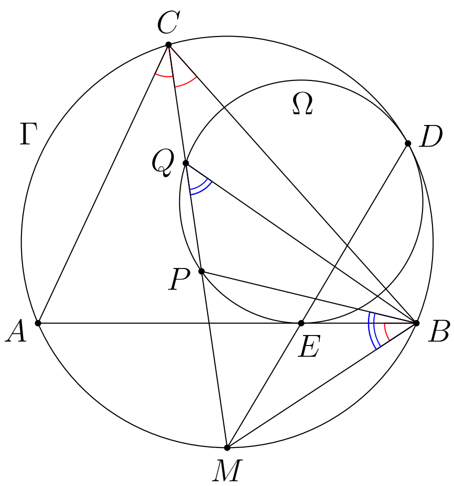
$$\frac{BC}{CP} = \frac{BA}{AE} = \frac{BA}{AD}. \tag{3}$$

Combining (2), (3) and (1), we find that

$$NQ \times NM = NM^2 \times \frac{CP}{CM} = \frac{NM^2}{CM} \times BC \times \frac{AD}{BA} = \frac{NM^2}{CM} \times \frac{BC}{BA} \times MA \times \frac{NC}{MN}.$$

Using $NM = \frac{1}{2}BA$, $CM = MA$ and $BC = 2NC$, this implies $NQ \times NM = NC^2$. The result follows readily. $\square$

Problem 7. (EGMO 2018 Problem 5) Let Γ be the circumcircle of triangle ABC. A circle Ω is tangent to the line segment AB and is tangent to Γ at a point lying on the same side of the line AB as C. The angle bisector of $\angle BCA$ intersects Ω at two different points P and Q. Prove that $\angle ABP = \angle QBC$.



Solution. WLOG assume Q lies between C and P. Suppose Γ and Ω touch each other at D, and Ω touches AB at E. Also, let M be the midpoint of arc AB not containing C. Then M lies on CP. In addition, M, E, D are collinear by homothety. Since M is the midpoint of arc AB, we have

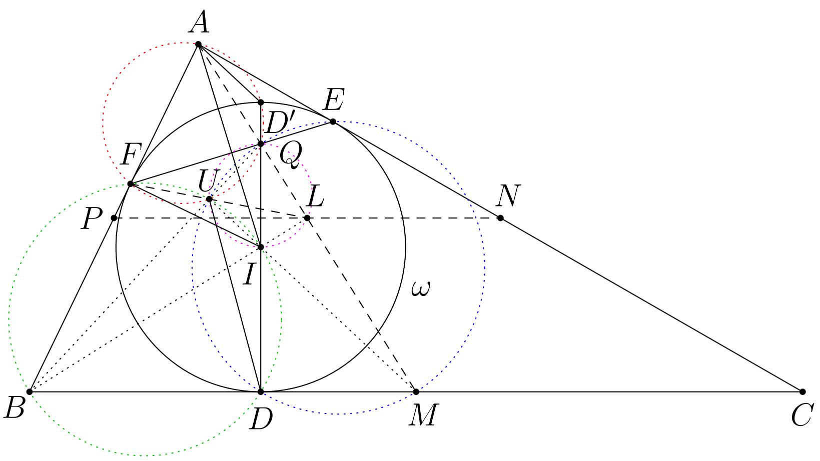
$$MB^2 = ME \times MD = MP \times MQ.$$

Thus, MB is tangent to (BPQ), and hence $\angle PBM = \angle PQB$. Thus, we find that

$$\angle PBA = \angle PBM - \angle ABM = \angle PQB - \angle ACM = \angle PQB - \angle MCB = \angle CBQ.$$

This completes the proof. $\square$

Problem 8. Let ABC be a triangle, I its incentre and ω its incircle. Let D, E and F be the points of tangency of ω with BC, AC and AB, respectively and M, N and P be the midpoints of BC, AC and AB. Let D' be the second intersection of DI with ω, Q the intersection of DI with EF and $U \neq Q$ be the intersection of $(AD'Q)$ with (DMQ). Suppose that U lies on the circumcircle of BDF. Prove that PN, AM, UF concur.



Solution. It is clear that PN passes through the midpoint L of AM. Therefore, it is the same as proving F, U, L are collinear. It is well-known that Q lies on AM. Since U lies on (BDF) and (MDQ), it also lies on (AFQ) (Miquel point). In other words, A, F, U, Q, D' are concyclic. Also, B, D, I, U, F are concyclic since $\angle IDB = \angle IFB = 90^\circ$.

Next, it is also well-known that $AD' // IM$. Therefore, we have $\angle MID' = \angle AD'I$, and hence

$$\angle IMB = 90^\circ - \angle DIM = \angle MID' - 90^\circ = \angle AD'I - 90^\circ = 90^\circ - \angle AFQ = \angle IAB.$$

Together with $\angle MBI = \angle ABI$ and $BI = BI$, we have $\triangle IBM \cong \triangle IBA$. Therefore, $BM = BA$, and BI is the perpendicular bisector of AM. It follows that B, I, L are collinear.

From $BM = BA$, we obtain $BF \times BA = BD \times BM$, and so A, F, D, M are concyclic. Considering the radical axes of $(AFUQ)$, $(DMQU)$ and $(AFDM)$, we find that B, U, Q are collinear.

Now, as $BI \perp QM$ and $QI \perp BM$, point I is the orthocentre of $\triangle QBM$. This shows $MI \perp BQ$. As we also have $\angle IUB = \angle IFB = 90°$, the points M, I, U are collinear. Thus, Q, U, I, L are concyclic, and

$$\angle QUL = \angle QIL = \angle BID = \angle BUF.$$

The last equality holds since $BD = BF$. This clearly implies F, U, L are collinear as desired. $\square$